

Unit Design I: Nitration Reactor Report

CENG0016: Process Plant Design Project

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1. Specification

(Company logo)		Reactor			
		Specification Sheet			
Reactor Name					
Equipment name & tag			R4: Nitration of Benzene		
N° of Units			R4		
Catalyst			Sulfuric Acid		
Type			CSTR		
Impeller (Y/N)			Y		
No of beds			0		
Bed Height (mm)					
			Bed 1		
			Bed 2		
			Bed 3		
			Bed 4		
Catalyst Specification					
Composition			0.586		
Geometrical Characteristics			D=1m, L=3m		
Catalyst Supplier			D-BASF		
Reactions					
(1)	$C_6H_6 + HNO_3 + H_2SO_4 \rightarrow C_6H_5NO_2 + H_2SO_4 + H_2O$				
(2)					
(3)					
(4)					
(5)					
(6)					
(7)					
(8)					
(9)					
(10)					
Reactor Parameters					
	Inlet	Outlet	Reactor Volume (m3)		
Flowrate (kmol/h)	425	313	Diameter (1000mm)		
Composition (mol%)			Height (3000mm)		
Benzene	51.00%	21.00%	No of Tubes (If applicable)		
Nitric acid	17.60%	9.00%	Length (m)		
Sulfuric acid	28.00%	12.00%	Diameter (mm)		
Nitrobenzene	0.00%	49.00%			
Water	3%	9%			
Temperature (°C)	25	78	Notes This unit design report explores the benzene nitration reaction under specified conditions, aiming for optimal results of 70% conversion at 351K with a composition of 58.6% sulfuric acid and 37.4% nitric acid. Through equation modeling and parameter studies, the report establishes the reactor design and operating parameters within a 1 meter diameter and 3 meter height cylinder CSTR reactor, incorporating a liquid-liquid plate heat exchanger. By achieving steady-state conditions and balancing mole and energy conversions, the report determines reactor volume and heat transfer area as functions of temperature. Sensitivity analyses reveal the model's dependency on sulfuric acid composition, leading to the identification of optimal conditions. Furthermore, the report addresses operational hazards associated with the use of corrosive sulfuric acid, emphasizing the importance of mitigating risks to workers, the community, and the environment during the reactor's		
Pressure (Barg)	1	1			
Press. Drop (Bar)	0				
Reactor Type	adiabatic	With plate heat exchange			
Insulation (v/n)					
Insulation type			Date 		
Insulation Thickness (mm)		15			
Heat Duty (MW)		18			
Heat Ex. Strategy		3987 W/m ² K			
Cooling/Heating media		water			
Cooling/Heating T(°C)		20	Signature 		
Cooling/Heating media Flowrate (kg/h)		1000			

2. Summary

This unit design report explores the benzene nitration reaction under specified conditions, aiming for optimal results of 70% conversion at 351K with a composition of 58.6% sulfuric acid and 37.4% nitric acid. Through equation modeling and parameter studies, the report establishes the reactor design and operating parameters within a 1 meter diameter and 3 meter height cylinder CSTR reactor, incorporating a liquid-liquid plate heat exchanger. By achieving steady-state conditions and balancing mole and energy conversions, the report determines reactor volume and heat transfer area as functions of temperature. Sensitivity analyses reveal the model's dependency on sulfuric acid composition, leading to the identification of optimal conditions. Furthermore, the report addresses operational hazards associated with the use of corrosive sulfuric acid, emphasizing the importance of mitigating risks to workers, the community, and the environment during the reactor's lifetime and disposal.

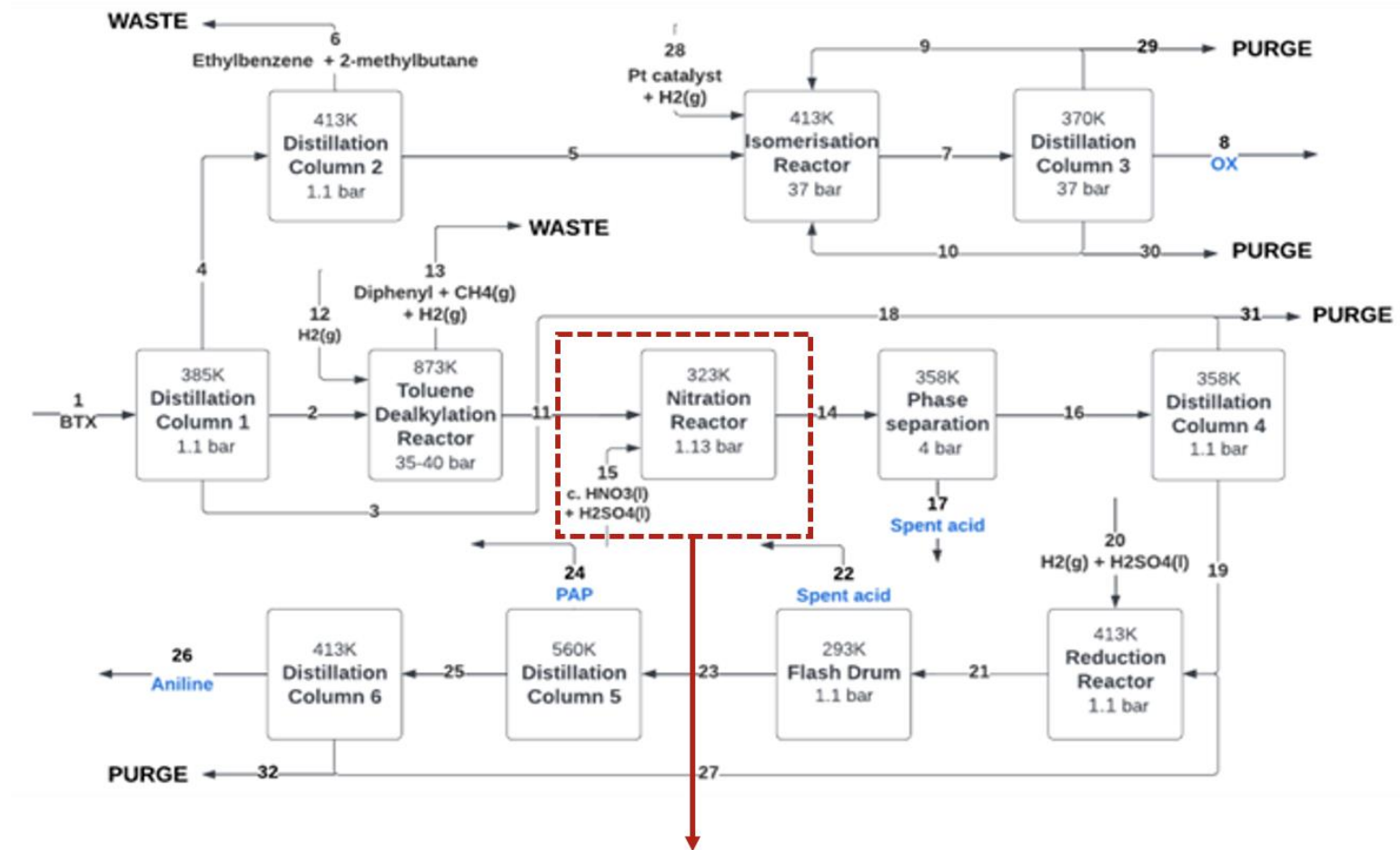
3. Introduction

During plant design, the feedstock Benzene 136.9 kmol/h , Toluene 314.8 kmol/h , Ortho-xylene 74.1 kmol/h , Meta-xylene 13.15 kmol/h , and Para-xylene 0.0539 kmol/h are used to produce Aniline 32.85 kmol/h , 4-Aminophenol 136.8 kmol/h , and increase Ortho-xylene purity from 0.1374 to 0.9978. In the designed plant as shown in Figure 3.2, there are two process routes to produce the final products Ortho-xylene and 4-Aminophenol respectively. For the first process route, by using isomerization reaction Metaxylene & Paraxylene $\rightarrow$ Orthoxylene, it purifies Ortho-xylene concentration from 0.1374 to 0.9978 in the end. For the second process route, it involves with 3 main reactions which are Toluene Dealkylation reaction $C_7H_8 + H_2 \rightarrow C_6H_6 + CH_4$ to convert the most Toluene to Benzene, Nitration reaction $C_6H_6 + HNO_3 + H_2SO_4 \rightarrow C_6H_5NO_2 + H_2SO_4 + H_2O$ to prepare the intermediate Nitrobenzene as the reactant of next reaction, and Reduction reaction $C_6H_5NO_2 + 3H_2 \rightarrow C_6H_5NH_2 + 2H_2O$ & $C_6H_5NO_2 + 2H_2 \rightarrow H_2NC_6H_4OH + H_2O$ to have main product 4- Aminophenol with side product Aniline. In this report, Nitration reaction of Benzene to produce Nitrobenzene reactor's unit design will be specified.

In nitration reactor, nitric acid HNO_3 will accept a hydrogen proton from sulfuric acid H_2SO_4 to form water and nitronium ions $H_2NO_3^+$ where it reacts as an electrophile with benzene ring C_6H_6 to form arenium ions, consequently, the arenium ion will lose its proton a hydrogen atom to $H_1SO_4^-$ what can form nitrobenzene and sulfuric acid H_2SO_4 in the end, in this case, the reaction can be written as $C_6H_6 + HNO_3 + H_2SO_4 \rightarrow C_6H_5NO_2 + H_2SO_4 + H_2O$ where sulfuric acid plays as a catalyst during this reaction. In the reaction, its properties like components phases, sensitivity, heat effect, kinetics, and catalyst can be discussed more. Under a atmosphere pressure, the limit of boiling point's range in the reaction is 80°C (benzene), in this case, by setting the reaction range is from 25°C to 80°C , all reactants and products are in liquid phase what can indicate this reaction is a homogenous reaction. Meanwhile, according to the S.M. Danov's research on *Kinetic of Benzene Nitration by Nitric Acid*, it uncovers that the reaction rate constant is high positively sensitive with the reaction temperature what indicate a linear relationship between the reaction rate constant and the reaction temperature, at the same time, the reaction is a exothermic reaction which increase the difficulties on control temperature to stay its sensitivity. For the catalyst, sulfuric acid involves as catalyst to offer protons to nitric acid to form nitronium ion, however, a few alternative solid catalysts like silica supported ammonium salt of 12-molybdophosphoric acid [1] and quinoline-2-formic acid with molybdovanadophosphoric heteropoly acid (QA-HMPV) [2] are indicated to be highly efficient and practical what can boost the yield to be 75% at 60°C , and the reaction still homogenous even the alternative solid catalysts are used since the operation conditions remain same.

Bases on the reaction itself properties, a Continuous Stirred-tank Reactor (CSTR) can be more suitable for benzene nitration reaction by comparing with Plug Flow Reactor (PFR). Due to the CSTR excellent mixing, temperature-control, and heat dissipation capacities, this homogenous, temperature-sensitive, and exothermic reaction can be operated better than PFR to avoid potential dead zones and hot spots under the reaction conditions 1 atm and 0°C to 80°C. From engineering and economical prospective, building up a CSTR is easier than a PFR what requires more complicated design like fixed catalyst bed consideration, in this case, the simpler design CSTR can bring some engineering and economical benefits like easy maintenance, meanwhile, due to the design of CSTR, controlling the reaction system can be more handy to change the conditions like reaction temperature, space time, reactor size, etc. which means CSTR can giving more redundancy to the whole plant system to adjust different working conditions.

3.2 Flowsheet



The reactor is specified at the below Figure 3.3

Figure 3.2: Block Flow Diagram to Produce 4-Aminophenol & Ortho-xylene

3.3 Block Diagram Nitration Reactor

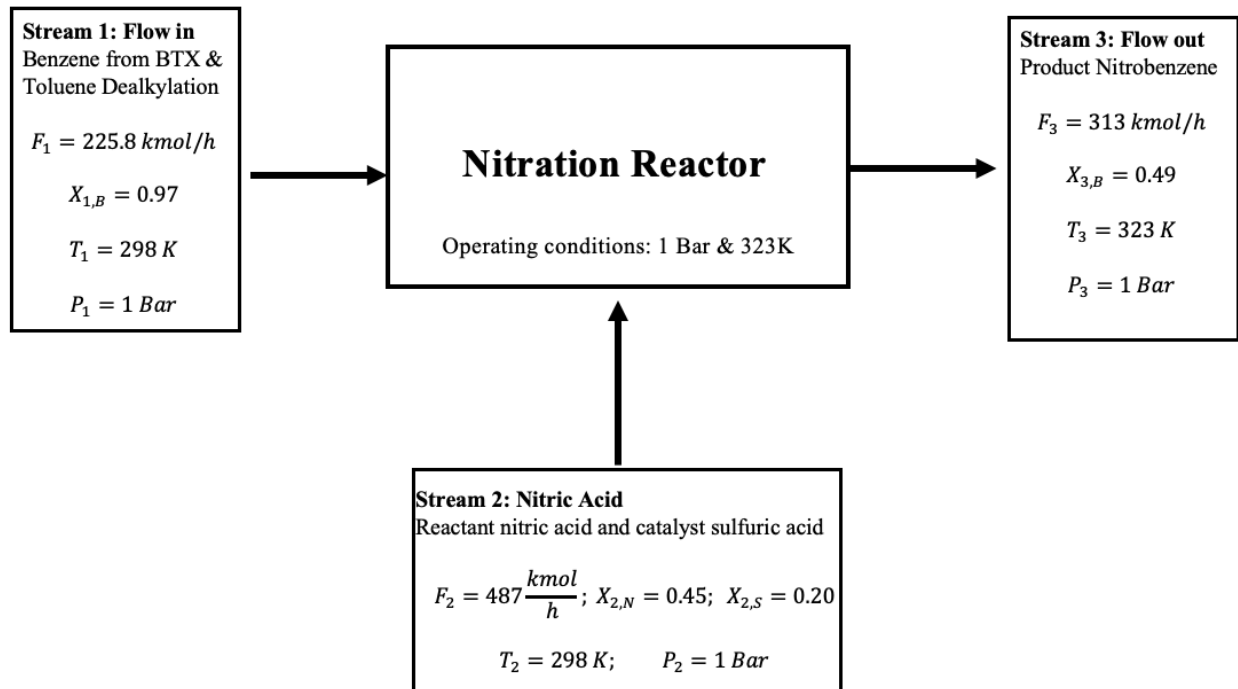
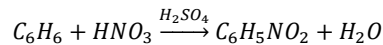


Figure 3.3: Specifications of Nitration Benzene Reactor

4 Design Methodology

4.1 Assumptions and Variables



1. This reactor operation is non-isothermal under **steady state**.
2. Due to nature of reaction, the power law model of rate can be defined as $-r_B = k * C_B * C_N$ which is a pseudo-second order reaction, however, according to the nature of benzene that low solubility in the aqueous phase, this reaction can be considered as a **pseudo-first order reaction**. [3]
3. The CSTR is **well mixed**, the reactor vessel's temperature and compositions are **uniform**.
4. This applied heat exchanger is under **steady state and coolant temperature at 293K**.
5. The **shaft work** from the surrounding and agitation in the system is negligible for analyzing.
6. There are closed boundaries to do mass transfer analysis.
7. The reaction is irreversible.

Meanwhile, all variables definitions are at **Appendix 4.1**.

4.2 Reaction kinetics

For benzene nitration reaction, it has been used widely in chemical industry, so its kinetics has been determined well. By the definition of rate constant,

$$k(T_r) = A * \exp\left(-\frac{E}{R * T_r}\right)$$

for an original temperature. However, due to the found research, the temperature is specified with its corresponding activation energy, a more specific expression to show its constant value k at any temperature can be represented by

$$k(T) = k(T_r) * \exp \left(-\frac{E}{R} * \left(\frac{1}{T} - \frac{1}{T_r} \right) \right)$$

From S.M. Danov's research, the rate constant of benzene's nitration with nitric acid has been calculated with Benzene's weight percentage, temperature, and activation energy.[1] According to the information, the expression can be formed as

$$k(T) = 0.000325 \text{ s}^{-1} * \exp \left(-\frac{75730 \frac{\text{J}}{\text{mol}}}{8.314 \frac{\text{J}}{\text{mol} * \text{K}}} * \left(\frac{1}{T} - \frac{1}{323\text{K}} \right) \right)$$

for 39% for HNO₃. However, here the mass and heat transfer can effect the value of rate constant. For mass transfer's potential effects, due to non-uniform distribution of input components like nitric acid, the kinetic is built bases on 39% Nitric acid what cannot be stable. For heat transfer's potential effects, since benzene nitration is an exothermic reaction which means the heat is not uniform in reactor fluid, the constant also will be unstable for this reason. For this report's discussion, the rate constant is defined as $k(T) = 0.000325 * \exp \left(-\frac{75730 \frac{\text{J}}{\text{mol}}}{8.314 \frac{\text{J}}{\text{mol} * \text{K}}} * \left(\frac{1}{T} - \frac{1}{323\text{K}} \right) \right)$ under 39% Nitric acid conditions at 323K.[4]

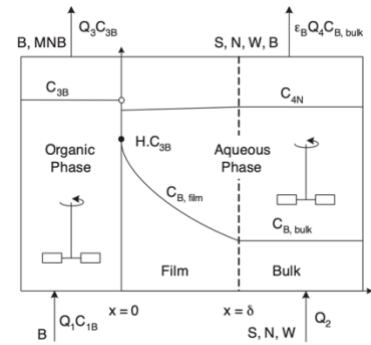
4.3 1-D Model

The mass transfer resistance analysis is made by finding out the potential limitation of the reaction (organic phase to aqueous phase). To do so, a one-dimensional film model is made to make further discussion[5]

$$F_B = -D1 \frac{dC_{B, \text{film}}}{dx} = k_L * \left(H * C_{3, B} * \cosh \left(Ha - X * \sqrt{\frac{k}{D1}} \right) - C_{B, \text{bulk}} * \left(X * \sqrt{\frac{k}{D1}} \right) \right) * \frac{Ha}{\sinh(Ha)}$$

According to Quadros result, with a very slow stirring speed 100 rpm, the reaction rate constant is only 18% lower than 350 rpm result. Moreover, by considering the activation energy $75730 \frac{\text{J}}{\text{mol}}$, it indicates a low activation

energy by comparing with other reaction which means nitric ions are moving voluntarily so that the kinetic energy is enough already. The phase change between organic and aqueous can be happened quickly so the limitation of mass transfer or the resistance of mass transfer can be treated as no influence to the system



4.4 Design equations

During the methodology, there are three ways to design a CSTR reactor by using respective three variables conversion, temperature, or volume. However, during this report, the aim is to discuss the best operating conditions to find optimal production process, since the range of reaction temperature is limited from 298K to 353K to avoid side reaction. Since this reaction is exothermic one, a cooler is needed to main the best working condition so that the conversion equations base on energy balance with a heat exchanger, meanwhile, a mole balance based conversion as well. According to the assumption and nature of CSTR reactor, these two conversions need to be equal to meet steady state definition. The following is going to bring up these two equations and do degree of freedom analysis to find extra information.

4.4.1 Energy balance conversion

Energy balance conversions [6],

$$X_{EB} = \frac{Cp_0 * (1 + \kappa) * (T - T_c)}{-\Delta H_{RX}^{\circ}}$$

Where κ and T_c are non-adiabatic parameters:

$$\kappa = \frac{U * A}{F_{B0} * Cp_0} \text{ and } T_c = (\kappa * T_a + T_0)/(1 + \kappa)$$

For $-\Delta H^\circ_{RX}$ the standard enthalpy of reaction,

$$-\Delta H^\circ_{RX} = \sum \Delta H^\circ_{products} - \sum \Delta H^\circ_{reactants}$$

According to the stoichiometry of reaction $C_6H_6 + HNO_3 \xrightarrow{H_2SO_4} C_6H_5NO_2 + H_2O$

$$-\Delta H^\circ_{RX}(298.3K) = -66800 \frac{J}{mol} \text{ (Appendix 4.4.1)}$$

For Cp_0 equals to $\sum \Theta_i * Cp_i$ the input mixture's heat capacity (benzene, nitric acid, water, and sulfuric acid),

$$Cp_0 = \sum \Theta_i * Cp_i = \frac{F_{B0}}{F_{B0}} * Cp_A + \frac{F_{N0}}{F_{B0}} * Cp_N + \frac{F_{W0}}{F_{B0}} * Cp_W + \frac{F_{S0}}{F_{B0}} * Cp_S = 188.18 \frac{J}{mol * K} \text{ (Appendix 4.4.1)}$$

For U the heat transfer coefficient, it is assumed by using a liquid-liquid plate heat exchanger, [7]

$$U = 3987 \frac{J}{s * K}$$

4.4.2 Mole balance conversion

Since this reaction's first order has been assumed, the mole balance conversion [8]

$$\text{First order: } X_{MB} = \frac{\tau * k}{1 + \tau * k} \text{ where } \tau = \frac{V}{v_0}$$

4.4.3 Degree of freedom

By making $X_{MB} = X_{EB}$ to meet the steady state condition, here is a degree of freedom to ensure the equations can be solved.

Unknown Variables: $X_{EB}(X_{MB})$, $V(L, D)$, $A(L, D)$

$$\text{Equations: } X_{EB} = \frac{Cp_0 * (1 + \kappa) * (T - T_c)}{-\Delta H^\circ_{RX}} = X_{MB} = \frac{\tau * k}{1 + \tau * k}$$

so the DOF is 2 which cannot be solved unless there are two more equations

Since the volume V of cylinder is also varied with length L and diameter D , meanwhile, the contact area A is defined by plate heat exchanger as the surface area of cylinder. In industrial solution, L/D ratio is selected between 2 to 3, however, due to all component's phase is liquid and the viscosity is not high, the ratio is recommended to be 3, so $\frac{L}{D} = 3$.[9]

$$V = \pi * r^2 * L = \pi * \frac{D^2}{4} * L = \frac{3}{4} * \pi * D^3$$

$$A = \pi * D * L + 2 * \pi * \frac{D^2}{4} = 3.5 * \pi * D^2$$

Now, the degree of freedom analysis has 3 unknown variables V, A, D , and 2 equations which means there is a variable need to be defined. Here, diameter D is assumed to be 1 meter, the value of it will be investigated later in 4.5 parameter study.

4.5 Sensitivity

4.5.1 Define Objectives and Identify Parameters

Here are two equations to find the focusing variables conversion X and reactor volume V . In order to find the optimal conversion and its corresponding reactor volume, the first thing is defining these two equations' respective objectives and parameters. For objectives, in this case, the input flow and temperature of benzene and the nature of this nitration reaction like standard enthalpy of formation are fixed since other units cannot be changed during this specific unit design. For parameters, these two equations need to be discussed separately here.

For energy balance conversion equation, $X_{EB} = \frac{Cp_0 * (1 + \kappa) * (T - T_c)}{-\Delta H_{RX}^\circ}$

In this equation, the parameters are κ and T_c non-adiabatic parameter, and Cp_0 input mixture heat capacity, however, for these 3 parameters can be defined by value of U heat transfer coefficient and Cp_0 mixture heat capacity, T_a coolant fluid temperature (steady state) and κ , and F_{N0} and F_{S0} respectively bases on flowing definitions:

$$\kappa(U * A, Cp_0) = \frac{U * A(D)}{F_{B0} * Cp_0} \text{ and } T_c(\kappa) = \frac{\kappa * T_a + T_0}{1 + \kappa}$$

$$Cp_0(F_{N0}, F_{S0}) = \sum \Theta_i * Cp_i = \frac{F_{B0}}{F_{B0}} * Cp_A + \frac{F_{N0}}{F_{B0}} * Cp_N + \frac{F_{W0}}{F_{B0}} * Cp_W + \frac{F_{S0}}{F_{B0}} * Cp_S$$

Objective: F_{B0} , T_0 , $-\Delta H_{RX}^\circ$

Parameters: U , F_{N0} , F_{S0} , D

Variable: T

In this case, there are 5 parameters can be tested later to find the maximum conversion;

For mole balance conversion equation, $X_{MB} = \frac{\tau * k}{1 + \tau * k}$

In this equation, the parameters are resistance time τ and reaction rate k . For resistance time, it can be defined by both the reactor diameter. In deeper definition of above values, they can be illustrated as

$$\tau = \frac{V(D)}{v_0}$$

$$k(T, T_r) = k(T_r) * \exp\left(-\frac{E}{R} * \left(\frac{1}{T} - \frac{1}{T_r}\right)\right), T_r(F_{N0}, F_{S0})$$

Objectives: F_{B0} , T_r ,

Parameters: F_{N0} , F_{S0} , D

Variables: T

In total, the parameters are U , F_{N0} , F_{S0} , and D what can be discussed to find its optimal solution.

4.5.2 Determine Ranges of Parameters

For U the heat transfer coefficient, its range is depend on material of coolant fluid and type of heat exchanger, so there is no specific range of it. However, to discuss its fluence to profiles, there are three values can be selected bases on three respective types: $U_1 = 3987 \frac{J}{s * K}$ (Plate HEX), $U_2 = 750 \frac{J}{s * K}$ (Air - cooled HEX), $U_3 = 1500 \frac{J}{s * K}$ (Plate HEX).

For F_{N0} , it is defined by specific nitric acid molar fraction, so its range can be from 0% to 100%, however, its value should be considered as not more than 68% since the maximum fraction value from the experiment data is 67.7%. [4] Same for F_{S0} , its range 0% to 88% is defined by the data

For D , from the rule of thumb for Equipment design, its range can be 0.1m to 3.9m

Since the ranges are specified, a proper method need to be investigated.

4.5.3 Method and Possible Judgement

To test the sensitivity, one-at-a-time (OAT) method is suitable to do. To do so, a parameter will be tested by three different values (one is original) and other parameters value stay constant, in the end, there will be 4 figures with 12 data sets to show the sensitivity. At this place, a few judgements can be made before found results. For the heat transfer coefficient, as it decreases, the non-adiabatic parameter κ decreases, the slope of X_{EB} decreases, the steady state point with X_{MB} getting higher (beyond first estimate temperature and conversion). For F_{N0} and F_{S0} , the increase in value of F_{S0} will bring lower rate constant and lower X_{MB} conversion, and the increase in value of F_{N0} will bring slightly lower rate constant. Finally, with the diameter, the increase in diameter will bring higher both conversions, but the increase in X_{EB} is larger than the increase in X_{MB} .

5 Results and Discussion

5.1 Model Implementation in Excel and Polymath

By finding the steady state point that the energy balance conversion equals to the mole balance conversion, by considering these equations are non-linear, a series of reaction temperature T is generated from 293K to 393K to find each temperature's respective MB and EB conversions by calculating every variable κ , T_c , τ , and k . However, the generated results only indicate two rough intersection points, to find the exact value of steady state points, Polymath non-linear equation solver can be used and the initial guesses is from Excel rough values, by changing the initial guesses of two steady points, the exact values can be found, and all relevant Excel calculations and Polymath code and results are at Appendix 5.1.

5.2 Conversion, Temperature, and Selectivity Profiles

By setting each T is increased 5K from 293K till 393K, 20 data sets are created with including temperature, mole balance conversion and energy balance conversion, and these data are presented at Figure 5.2.

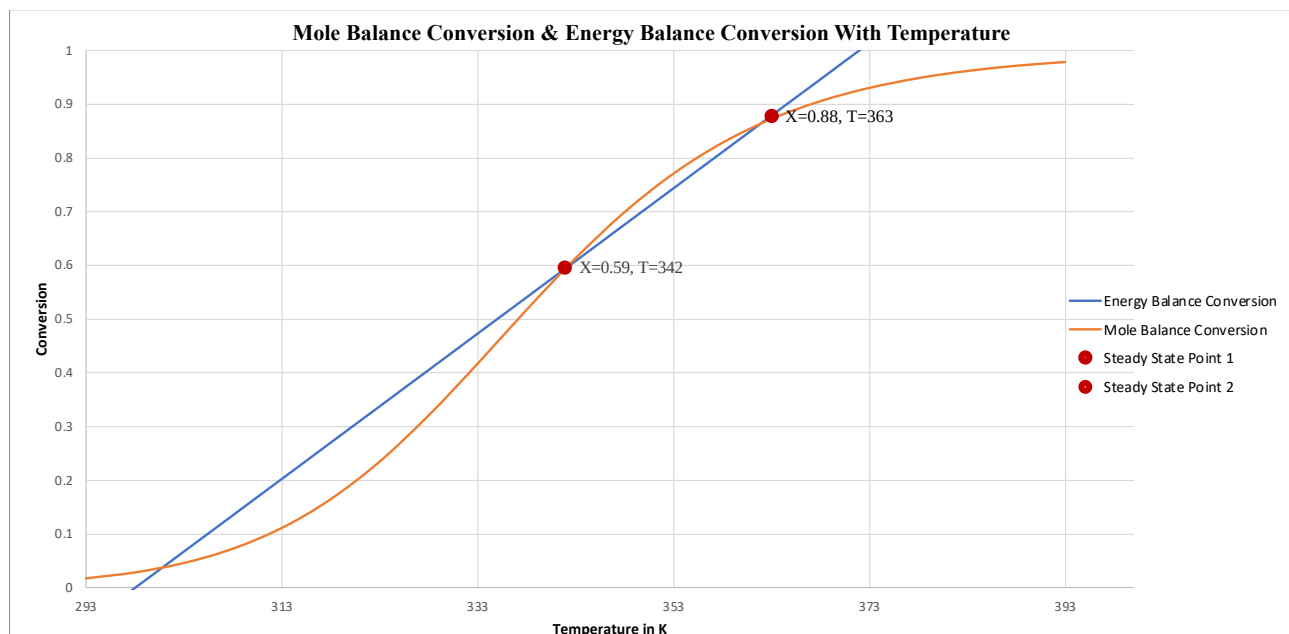


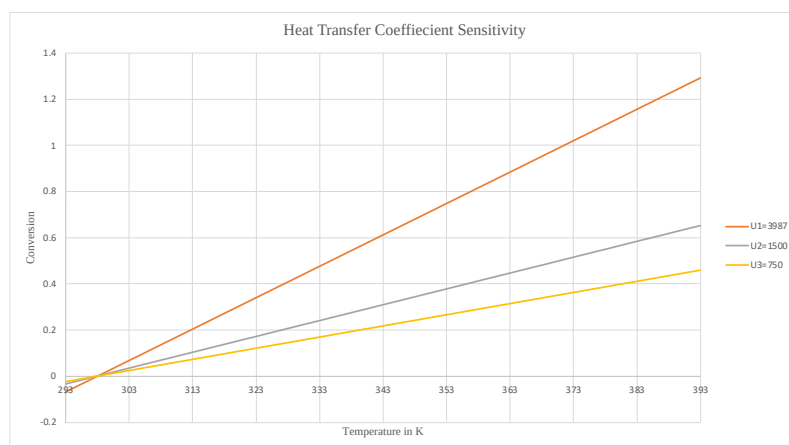
Figure 5.2: Mole Balance and Energy Balance Conversions with Specific Steady State Points

By specifying steady state points and highlight them, the Figure 5.2 is created what shows the mole balance function has two different slope trends that a increase one and a decrease one since the exponential function of temperature, and the energy balance function is a linear one with constant slope from 293K the defined non-adiabatic coefficient temperature to 365K (limited by conversion). Bases the concept of conversion, the higher energy can bring more kinetic energy to components to share electrons and molecular what can increase conversion eventually, so both conversion functions have positive slope. For mole balance conversion equations, the less thermal energy as the denominator of exponential term can bring faster growth slope but with low conversion value, however, as more and more thermal energy enters the system, the conversion is getting higher but with slower slope which means the component's kinetic energy is getter maximum.

Meanwhile, the figure also shows the steady state will appear when reactor temperature is 337K and 357K to have respectively 0.6 and 0.87 conversion. However, for the second steady state point, it is beyond the range of reaction temperature what would bring side reaction to have impurity, but there is a trade-off can be conducted between separation of impurities or building series CSTR reactor by considering the second steady state point's conversion is much higher than the first one.

5.3 Parameter Studies

From parameter studies, the Appendix 5.3 is settled with different values of heat transfer coefficient, composition of nitric acid, composition of sulfuric acid, and diameter of reactor. The following graphs can illustrate the sensitivities well.



For heat transfer coefficient, here are 3987, 1500, and 750 watt per square meter per K. By changing its value, only energy balance conversion result changes. According to the Figure 5.3.1, the higher heat transfer coefficient will bring higher function slope what indicate the better heat exchanger used, the better performance the reactor will have.

Figure 5.3.1: Heat Transfer Coefficient Sensitivity Test

For composition of nitric acid, by considering the below Figure 5.3.2.a and 5.3.2.b with Figure 5.2, the higher composition nitric acid can bring both two steady state point lower which makes sense because nitric acid brings more N_2O ions to the system. However, by comparing Figure 5.3.2.b with Figure 5.2, the lower steady state points also means the steady state conversion is not ideal enough, so its sensitivity to the system is not high as the heat coefficient.

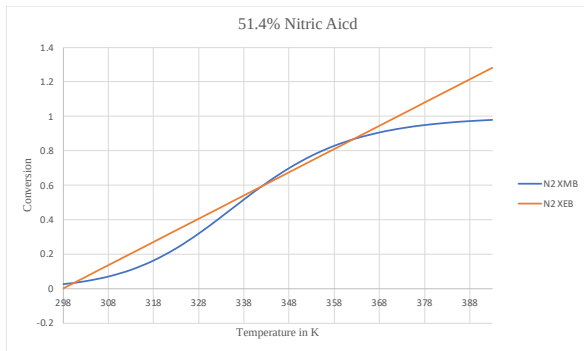


Figure 5.3.2.a: 51.4% Nitric Acid Sensitivity Test

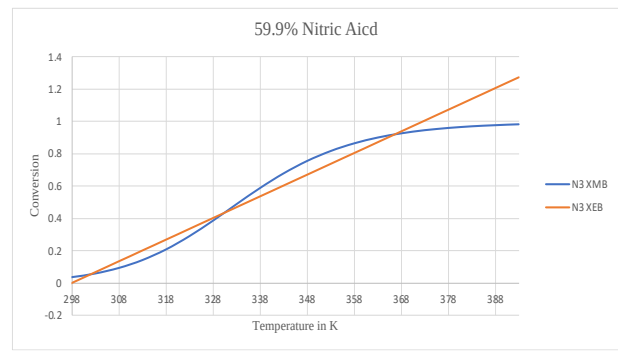


Figure 5.3.2.b: 59.9% Nitric Acid Sensitivity Test

For composition of sulfuric acid, by considering the below Figure 5.3.3.a and 5.3.3.b with Figure 5.2, the higher sulfuric acid composition can make the mole balance conversion slope higher since the reference rate constant is increasing. Bases on the Figure 5.3.3.b, it indicates a steady state around 353K with 0.75 conversion which is providing a better reaction performance than Figure 5.2 case. By comparing the sensitivities of nitric acid composition and sulfuric acid composition, the sulfuric acid as a catalyst defines whole reaction's kinetics, which indicates the reaction is more sensitive with composition of sulfuric acid.

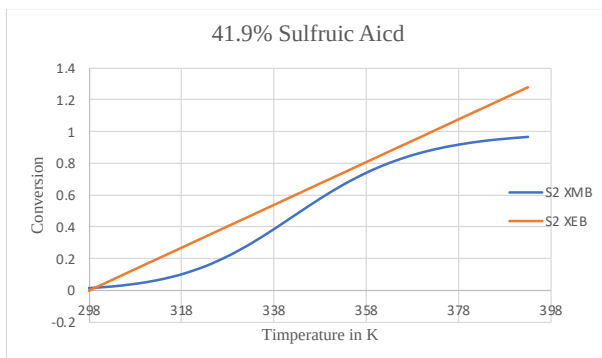


Figure 5.3.3.a: 41.9% Sulfuric Acid Sensitivity Test

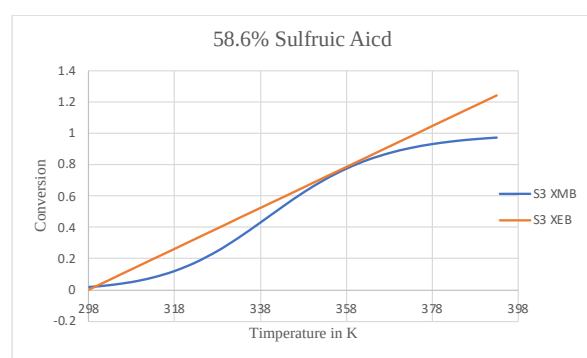


Figure 5.3.3.b: 58.6% Sulfuric Acid Sensitivity Test

At the last, the diameter of CSTR reactor, to make judgements on this parameter, the ratio of length and diameter also will need to be discussed. by considering the below Figure 5.3.4.a and 5.3.4.b with Figure 5.2, the increase of diameter brings huge change in both mole balance conversion and energy balance conversion. For mole balance conversion, a higher diameter value brings higher function slope by increasing the contact area, at 338K, both data sets show a over 1 conversion. For energy balance conversion, it illustrates at the same way, however, the increase in diameter undermines exponential term influence, so the increase in temperature as the denominator can bring a higher slope. By considering the ratio of length and diameter, it indicates that a higher ratio brings more advantages to mole balance conversion, eventually, a higher ratio can bring steady state points to be lower. Bringing back to the last mythology part, the diameter is made to be 1 meter which is fair enough since the purpose is meeting the steady state with a high conversion that a larger diameter cannot do. The reason not choosing a lower value is based on construction consideration, too short reactor diameter means that there are rare reactions can also react inside this design reactor so that the function of this CSTR is too narrow.

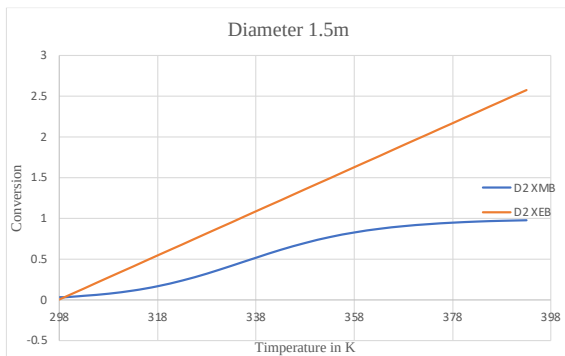


Figure 5.3.4.a: Diameter 1.5m Sensitivity Test

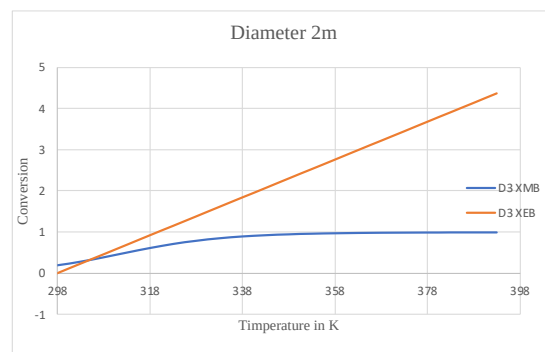


Figure 5.3.4.b: Diameter 2m Sensitivity Test

To find optimal production conversion, bases on found solutions, a optimal result is calculated

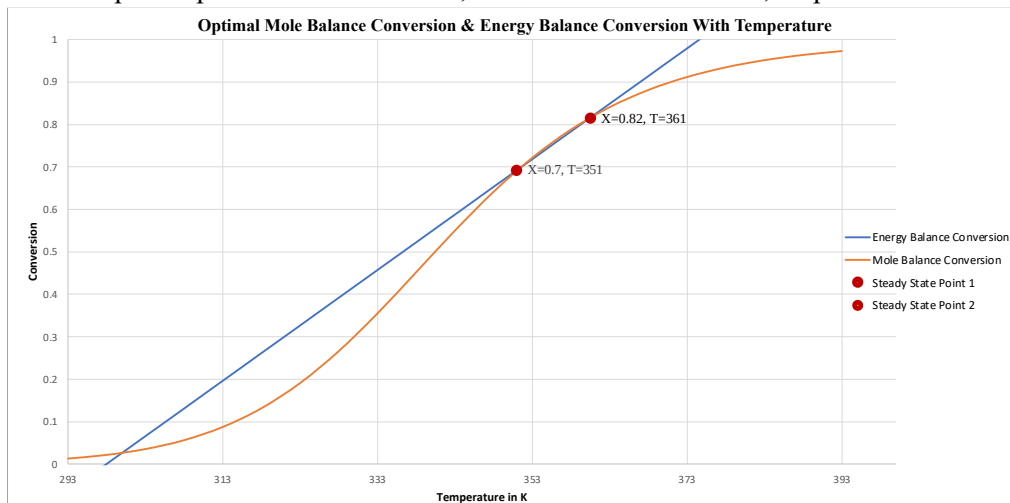


Figure 5.3.5: Optimal Mole Balance and Energy Balance Conversions with Specific Steady State Points

By using 58.6% Sulfuric Acid, 37.4% Nitric Acid, 1 m diameter, and 3987 watt per K per square meter, a optimal is generated to have two steady state points 70% conversion at 351K and 82% conversion at 361K by using Appendix 5.3.5 code. For this result, its steady state point is moving forward by comparing with the original result and the conversion is much higher which is desirable. From the research of R.C. Miller and Paulo A. Quadors, the experimental data indicates a 75% conversion at 343K with a much higher sulfuric acid composition 82%. By comparing with their result, the reason of difference is that the used catalyst is higher than the former research, so a lower activation with a high reaction rate constant is estimated. [4][5]

5.4 Safety Considerations

Equipment: high Temperature (sensor) lead to explosion, overloading (sensor and vessel) explosion, broken tank high corrosive components, dead space(agitator) less productive

During the process and surrounding the unit, due to the high corrosive catalyst sulfuric acid and reactant nitric acid, the safety is important to consider. Here are few topics to discuss include sensor issue, vessel issue, and agitator issue during the operation time what may hurt workers. For sensor issue, a very high reactor temperature may appear inside the reactor if the thermal sensor is broken, so that the components will trans phase from liquid to gas and it bring a high pressure inside the vessel so that it will explode. Meanwhile, the sensor of position of liquid level can misconduct operation, so the vessel may be overloaded so that a high inside pressure will be made to explode. For vessel issue, due to limited service life of all materials, the vessel

can be rusted or be broken by surrounding environment, a leaking will happen then what will bring workers, the community, and environment into a serious chemical danger. For agitator, if the rotation speed is not working effective enough, dead volume inside vessel will be created what will decrease productivities, moreover, the mixture temperature will not be uniform so a high temperature volume can be made that will also lead to phase change. At the end, nature disasters can also be an issue like earthquake or storm what requires high quality constructions. To prevent safety issues, safety training, regulated equipment inspection scheme, and high quality construction can be taken to help workers, the community, and environment avoid damages.

6. Environmental Reflection

For the sourcing, lifetime, and disposal of the catalyst, sulfuric acid as a strong corrosive acid needs to be considered carefully. For the sourcing of sulfuric acid, it can be from other chemical companies or self-made way since it is one of the most common and used chemical in chemical industry. Sulfuric acid can be made by oxidizing sulfur dioxide to sulfur trioxide what can be reacted with water, in the process, sulfur dioxide can be got from other reactions like fossil burning. For the lifetime aspect, sulfuric acid in the reaction as a catalyst can be recycled again from separation unit to save cost, meanwhile, it is high corrosive component like the last mentioned, any leaking of it will bring huge damages to workers, the local community, the surrounding environment, and equipment, so safety training, regulated equipment inspection scheme, and high quality construction need to be taken.

The most important part is that disposal of it since sulfuric acid may become spent or contaminated with impurities, rendering it unsuitable for further use as a catalyst. Spent acid may contain various toxic substances depending on the specific industrial process it was used in, such as heavy metals or organic compounds. Meanwhile, spent sulfuric acid often requires treatment before disposal to minimize environmental impact. Treatment methods may include neutralization to adjust pH, precipitation to remove contaminants, or chemical conversion to less hazardous forms. Lastly, disposal of spent sulfuric acid is subject to strict regulations aimed at protecting human health and the environment. Depending on the jurisdiction, disposal may require permits and compliance with specific handling and disposal protocols.

7. Conclusion

By discussing the order, kinetics, design questions, steady state, sizing, parameter of benzene nitration reaction with given input condition, this unit design report is made by building up the equation model and parameter studies to have the optimal result 70% conversion at 351K with 58.6% sulfuric acid and 37.4% nitric acid for inlet temperature 298K and given flowrate inside a 1 meter diameter and 3 meter height cylinder CSTR reactor with a liquid-liquid plate heat exchanger. By finding the steady state point, both mole balance and energy balance conversions are made to be equal with assuming an initial value of diameter, with sensible length and diameter ratio, the reactor volume and heat transfer contact area can be found to have both conversions function of temperature. By doing the parameter studies, this model is sensitive with the composition of catalyst sulfuric acid, by using estimated experimental data, an optimal result is found with considering no mass transfer resistance. In the end, there is discussion about potential HAZARD from operation and analysis of the catalyst the strong corrosive sulfuric acid to avoid damages to worker, the local community, the surrounding environment, and plant equipment in the lifetime and disposal aspects.

Appendix

4.1 Variables

k , reaction rate constant, s^{-1} ;

T , reaction temperature, K ;

T_r , reaction rate constant reference temperature, K ;

E , activation energy, $\frac{J}{mol \cdot K}$;

R , ideal gas constant, $8.314 \frac{mol}{J \cdot K}$;

X_{EB} , the energy balance conversion;

X_{MB} , the mole balance conversion;

Cp_0 , the mixture heat capacity, $\frac{J}{mol \cdot K}$;

κ and T_c , non – adiabatic coefficient ;

$-\Delta H^\circ_{RX}$, the standard enthalpy of formation;

U , the heat transfer coefficient, $\frac{W}{m^2 \cdot K}$;

T_a , inlet coolant temperature, K ;

τ , the resistance time, s ;

V , reactor volume, m^3 ;

A , the heat transfer contact area, m^2 ;

v_0 , inlet volumetric flowrate, m^3/s ;

B , benzene;

N , nitric acid;

NB , nitrobenzene;

S , sulfuric acid;

F_{i0} , the inlet flowrate of component i

4.2 Kinetic Expression

From the table below,

TABLE 1. EXPERIMENTAL NITRATIONS OF 1,3-BIS(TRIFLUOROMETHYL)BENZENE

Run No.	Temperature, °C	H ₂ SO ₄ Mole %	HNO ₃ Mole %	Density of acid (CF ₃) ₃	Moles HNO ₃ per Mole (CF ₃) ₃	k, Moles per Hour per Liter of acid	k Corrected to 50° C
1	50.0	35.8	39.0	1.74	6.33	1.165	1.17
2	50.0	40.7	1.0	1.82	0.24	0.000 ^b	0.00
3	50.0	39.7	35.3	1.73	5.65	0.737	0.737
4	50.0	51.3	48.2	1.78	6.02	0.960	0.960
5	50.0	54.0 ^a	43.0	1.82	5.32	0.753	0.753
6	50.0	44.6	51.6	1.74	6.67	1.428	1.43
7	50.5	39.5	51.5	1.72	6.67	1.322	1.32
8	50.0	41.7	43.9	1.74	5.72	0.860	0.860
9	50.0	34.0	44.7	1.71	5.88	0.983	0.983
10	50.0	49.1	34.6	1.74	5.92	0.655	0.655
11	50.0	37.7	51.4	1.71	6.10	1.170	1.17
12	65.5	46.7	33.1	1.77	6.13	2.621	0.728
13	45.0	46.7	33.1	1.77	6.13	0.450	0.704
14	60.0	46.7	33.1	1.77	6.13	1.893	0.822
15	50.0	30.5	54.4	1.69	5.88	1.432	1.43
18	60.5	63.9	30.0	1.82	5.47	1.760 ^b	0.733
19	60.5	69.6 ^a	23.5	1.87	5.47	1.731	0.721
20	60.5	58.6	37.4	1.81	5.49	1.977	0.825
21	60.5	70.1	14.6	1.84	5.88	1.850 ^b	0.771
22	65.5	74.7	7.3	1.84	2.93	4.960 ^b	1.38
26	65.5	74.9	15.4	1.85	5.21	2.420 ^b	0.677
27	65.5	74.9	15.4	1.85	5.21	2.970 ^b	0.828
28	65.5	88.2	8.0	1.83	2.93	6.740 ^b	1.88
29	65.5	66.9	6.5	1.83	2.93	3.630 ^b	1.01
30	65.5	80.4	7.7	1.85	2.93	5.820 ^b	1.62
31	65.5	59.2	12.2	1.81	5.88	2.150 ^b	0.600
32	65.5	64.0	22.0	1.83	5.52	2.800 ^b	0.782
33	65.5	48.7	10.2	1.76	2.98	1.049	0.293
34	65.0	0.0	72.8	1.49	11.9	0.000 ^c	0.00
34a	80.0	0.0	72.8	1.49	11.9	0.000 ^c	0.00
35	50.0	42.1	39.1	1.74	5.88	0.698	0.699
35a	60.0	42.1	39.1	1.74	5.88	2.060	0.897
36	74.3	48.9	10.1	1.76	3.97	1.694	0.240
37	65.0	54.3	21.2	1.78	5.92	2.048	0.593
38	57.5	29.8	64.3	1.68	9.48	3.339	.77
39	51.3	20.8	65.7	1.64	20.8	1.102	0.979
40	50.0	38.4	58.7	1.73	8.30	1.501	1.501
41	61.9	13.1	66.6	1.59	22.4	2.259	0.837
42	52.0	25.3	56.6	1.66	5.92	2.334	1.96
43	60.0	34.6	41.5	1.70	6.03	1.896	0.827
43R	60.2	34.6	41.5	1.70	6.03	1.960	0.839
44	69.0	45.9	19.6	1.75	5.85	1.463	0.308
46	59.5	6.1	67.7	1.53	24.2	0.152	0.0687
47	50.1	23.8	60.0	1.65	9.60	1.335	1.32
48	60.0	9.5	67.3	1.55	11.6	0.812	0.353
49	55.2	19.1	55.9	1.61	9.60	0.580	0.371
50	68.2	14.9	52.3	1.58	19.2	0.0688	0.0153
51	49.6	33.3	59.9	1.71	8.32	1.525	1.57
52	45.5	31.5	62.1	1.70	8.73	1.122	1.67
53	55.8	31.5	62.1	1.70	8.73	2.707	1.65
54	65.3	55.4	27.3	1.79	5.52	2.595	0.731
55	74.5	41.0	22.9	1.73	5.52	1.653	0.231
56	80.0	57.0	10.1	1.79	4.77	5.110	0.477
57	78.7	63.7	6.5	1.81	3.30	7.340	0.756
57R	73.3	63.7	6.5	1.81	1.65	5.340	0.812
58	76.6	33.0	23.9	1.68	11.0	0.198	0.0238
59 ^d	72.5	27.8	47.1	1.73	13.1	2.161	0.353
60 ^d	72.6	41.9	33.7	1.75	8.92	4.230	0.690

^a Third component SO₃ instead of H₂O. ^b Taken from slopes of curves for moles converted versus time. ^c Zeroes mean rate too slow for measurement. ^d From unpublished results of C. Shih.

4.4.1 Standard enthalpy of formation

$$\begin{aligned}\Delta H^{\circ}_{RX}(298.3K) &= (\Delta H^{\circ}_{NB} + \Delta H^{\circ}_{water}) - (\Delta H^{\circ}_B + \Delta H^{\circ}_N) \\ &= 61.5 \frac{kJ}{mol} - 285.83 \frac{kJ}{mol} - \left(-206.28 \frac{kJ}{mol} + 49.2 \frac{kJ}{mol} \right) = -66.8 \frac{kJ}{mol}\end{aligned}$$

Mixture heat capacity

$$\begin{aligned}C_{p0} &= \sum \Theta_i * C_{p_i} = \frac{F_{B0}}{F_{B0}} * C_{p_A} + \frac{F_{N0}}{F_{B0}} * C_{p_N} + \frac{F_{W0}}{F_{B0}} * C_{p_W} + \frac{F_{S0}}{F_{B0}} * C_{p_S} \\ &= 133 + \frac{21.7}{60.8} * 53.2 + \frac{19.9}{60.8} * 83.68 + \frac{1.89 + 14}{60.8} * 33.8 = 188.18 \frac{J}{mol} \text{ (Appendix 4.4.2)}\end{aligned}$$

5.1 Result

Parameters			
	Value	Unit	Descriptor
Cp0	188.18	J/mol K	inlet mixture heat capacity
Hrex	66800	J/mol	standard enthalpy of formation
U	3988	W/m^2 K	heat transfer coefficient
Ta	293	K	Coolant fluid temperature
T0	298	K	inlet temperature
v0	24.3	mol/L	inlet volumetric flowrate
E	75730	J/mol	activation energy
k0	0.00049	1/s	reference rate constant
FN0	21.7	mol/s	inlet nitric acid molar flowrate
FBO	60.8	mol/s	inlet benzene molar flowrate
FSO	19.9	mol/s	inlet sulfuric acid molar flowrate
Variables			
	value	Unit	Descriptor
D	1	m	reactor diameter
Tau	942.45	s	resistance time
V	2.356125	m^3	reactor volume
A	10.99525	m^2	reactor area
U*A	43838.0618	W/K	heat transfer
KA	3.83154826	dimensionless	non-adiabatic coefficient

T	XEB	k, rate constant	XMB
293	-0.0652432	1.393E-05	0.01295213
298	0	2.34684E-05	0.02162907
303	0.06524325	3.88632E-05	0.03531624
308	0.13048649	6.33113E-05	0.05628262
313	0.19572974	0.000101544	0.08730326
318	0.26097298	0.000160462	0.13130753
323	0.32621623	0.00025	0.19061109
328	0.39145947	0.00038427	0.26577605
333	0.45670272	0.000583077	0.35453012
338	0.52194596	0.000873893	0.4515161
343	0.58718921	0.001294396	0.54941179
348	0.65243246	0.001895717	0.64103223
353	0.7176757	0.002746537	0.72123402
358	0.78291895	0.00393822	0.78767692
363	0.84816219	0.005591171	0.84043105
368	0.91340544	0.007862662	0.88104624
373	0.97864868	0.010956374	0.91166783
378	1.04389193	0.015133939	0.93445275
383	1.10913517	0.020728807	0.95128257
388	1.17437842	0.028162774	0.96367511
393	1.23962166	0.037965556	0.97279915
351	0.6915784	0.002370998	0.69073569
360.5	0.81554057	0.004698175	0.81569132

5.2

X Function Parameter Study		Value	kA	Cp0	E	tau
U	1	3987	3.833203512	188.18	75730	942
	2	1500	1.442138266	188.18	75730	942
	3	750	0.721069133	188.18	75730	942
FN0	1	0.39	3.832620974	188.2086024	75730	942
	2	0.514	3.788341316	190.4084603	75730	942
	3	0.599	3.758574744	191.9164274	75730	942
FS0	1	0.358	3.832620974	188.2086024	75730	942
	2	0.419	3.776785482	190.9910532	75730	942
	3	0.586	3.631928833	198.6085824	75730	942
D	1	1	3.83	188.18	75730	942
	2	1.5	8.620983596	188.18	75730	3180
	3	2	15.32619306	188.18	75730	7540
Maximum		0.586	3.631928833	198.6085824	75730	942

5.3

U					
T	K	XMB	Xeb 1	Xeb 2	XEB 3
293	1.8109E-05	0.0167726	-0.0680321	-0.0343682	-0.0242268
298	3.05089E-05	0.02793652	0	0	0
303	5.05222E-05	0.04542978	0.06803214	0.0343682	0.02422677
308	8.23047E-05	0.0719525	0.13606428	0.06873641	0.04845353
313	0.000132007	0.11059757	0.20409642	0.10310461	0.0726803
318	0.000208601	0.16423039	0.27212856	0.13747281	0.09690707
323	0.000325	0.23439115	0.3401607	0.17184102	0.12113383
328	0.000499551	0.31999478	0.40819284	0.20620922	0.1453606
333	0.000758001	0.41658191	0.47622499	0.24057743	0.16958737
338	0.001136061	0.51694785	0.54425713	0.27494563	0.19381413
343	0.001682715	0.61317041	0.61228927	0.30931383	0.2180409
348	0.002464432	0.69893075	0.68032141	0.34368204	0.24226766
353	0.003570499	0.77082143	0.74835355	0.37805024	0.26649443
358	0.005119686	0.82825968	0.81638569	0.41241844	0.2907212
363	0.007268523	0.87256192	0.88441783	0.44678665	0.31494796
368	0.010221461	0.90591438	0.95244997	0.48115485	0.33917473
373	0.014243286	0.93063829	1.02048211	0.51552305	0.3634015
378	0.01967412	0.94880464	1.08851425	0.54989126	0.38762826
383	0.026947449	0.96209895	1.15654639	0.58425946	0.41185503
388	0.036611606	0.97182157	1.22457853	0.61862766	0.4360818
393	0.049355223	0.9789441	1.29261067	0.65299587	0.46030856

FN0						
T	K2	K3	XMB 2	XMB 3	Xeb 2	XEB 3
293	1.8109E-05	2.4294E-05	0.0167726	0.02237292	-0.0674687	-0.067046168
298	3.05089E-05	4.0929E-05	0.02793652	0.03712371	0	0
303	5.05222E-05	6.7777E-05	0.04542978	0.06001462	0.06746873	0.067046168
308	8.23047E-05	0.00011041	0.0719525	0.09421184	0.13493746	0.134092335
313	0.000132007	0.00017709	0.11059757	0.14297043	0.20240618	0.201138503
318	0.000208601	0.00027985	0.16423039	0.20861969	0.26987491	0.268184671
323	0.000325	0.000436	0.23439115	0.29113809	0.33734364	0.335230838
328	0.000499551	0.00067017	0.31999478	0.38699087	0.40481237	0.402277006
333	0.000758001	0.00101689	0.41658191	0.48925068	0.47228109	0.469323174
338	0.001136061	0.00152407	0.51694785	0.58943602	0.53974982	0.536369341
343	0.001682715	0.00225743	0.61317041	0.68015315	0.60721855	0.603415509
348	0.002464432	0.00330613	0.69893075	0.75694971	0.67468728	0.670461677
353	0.003570499	0.00478996	0.77082143	0.81858237	0.742156	0.737507844
358	0.005119686	0.00686826	0.82825968	0.86612935	0.80962473	0.804554012
363	0.007268523	0.009751	0.87256192	0.90182068	0.87709346	0.87160018
368	0.010221461	0.01371248	0.90591438	0.92814626	0.94456219	0.938646347
373	0.014243286	0.01910792	0.93063829	0.94736747	1.01203091	1.005692515
378	0.01967412	0.02639359	0.94880464	0.96133437	1.07949964	1.072738683
383	0.026947449	0.03615104	0.96209895	0.97147281	1.14696837	1.13978485
388	0.036611606	0.04911588	0.97182157	0.97884366	1.2144371	1.206831018
393	0.049355223	0.06621193	0.9789441	0.98422007	1.28190582	1.273877186

FS0						
T	K2	K3	XMB 2	XMB 3	Xeb 2	XEB 3
293	1.06983E-05	1.27599E-05	0.00997722	0.01187707	-0.0673279	-0.0652151
298	1.80237E-05	2.14971E-05	0.0166949	0.01984829	0	0
303	2.98469E-05	3.55987E-05	0.02734693	0.03244593	0.06732787	0.06521507
308	4.86231E-05	5.79932E-05	0.04379693	0.05179978	0.13465575	0.13043015
313	7.79856E-05	9.3014E-05	0.06843501	0.08056057	0.20198362	0.19564522
318	0.000123235	0.000146983	0.10401276	0.12161914	0.2693115	0.2608603
323	0.000192	0.000229	0.15316243	0.17744082	0.33663937	0.32607537
328	0.000295119	0.000351991	0.21752884	0.24901005	0.40396725	0.39129045
333	0.000447803	0.000534099	0.29668148	0.33471766	0.47129512	0.45650552
338	0.00067115	0.000800486	0.38733877	0.42989347	0.53862299	0.5217206
343	0.000994096	0.001185667	0.48358811	0.52761075	0.60595087	0.58693567
348	0.001455911	0.001736477	0.57832023	0.62060295	0.67327874	0.65215075
353	0.002109341	0.002515828	0.66521582	0.70325619	0.74060662	0.71736582
358	0.003024553	0.00360741	0.74020097	0.77263323	0.80793449	0.7825809
363	0.004294019	0.005121513	0.80178262	0.82831041	0.87526237	0.84779597
368	0.006038525	0.007202199	0.85048496	0.87153912	0.94259024	0.91301105
373	0.008414495	0.010036039	0.88797344	0.90434236	1.00991811	0.97822612
378	0.011622865	0.013862688	0.91630916	0.92886943	1.07724599	1.0434412
383	0.015919723	0.018987587	0.93748585	0.94705159	1.14457386	1.10865627
388	0.021629011	0.025797101	0.95321534	0.96047567	1.21190174	1.17387135
393	0.029157547	0.03477645	0.96487088	0.97037863	1.27922961	1.23908642

T	D					
	K1		XMB 2	XMB 3	Xeb 2	XEB 3
293	1.8109E-05		0.05445109	0.12013823	-0.1355009	-0.2295909
298	3.05089E-05		0.02793652	0.18701648	0	0
303	5.05222E-05		0.04542978	0.27585404	0.13550087	0.22959087
308	8.23047E-05		0.0719525	0.38293609	0.27100174	0.45918174
313	0.000132007		0.11059757	0.49883014	0.4065026	0.6887726
318	0.000208601		0.16423039	0.611326	0.54200347	0.91836347
323	0.000325		0.23439115	0.71018693	0.67750434	1.14795434
328	0.000499551		0.31999478	0.79020746	0.81300521	1.37754521
333	0.000758001		0.41658191	0.85108689	0.94850608	1.60713608
338	0.001136061		0.51694785	0.89546204	1.08400695	1.83672695
343	0.001682715		0.61317041	0.92694156	1.21950781	2.06631781
348	0.002464432		0.69893075	0.94893222	1.35500868	2.29590868
353	0.003570499		0.77082143	0.96418538	1.49050955	2.52549955
358	0.005119686		0.82825968	0.97474903	1.62601042	2.75509042
363	0.007268523		0.87256192	0.98208035	1.76151129	2.98468129
368	0.010221461		0.90591438	0.98719095	1.89701216	3.21427216
373	0.014243286		0.93063829	0.99077443	2.03251302	3.44386302
378	0.01967412		0.94880464	0.993304	2.16801389	3.67345389
383	0.026947449		0.96209895	0.99510245	2.30351476	3.90304476
388	0.036611606		0.97182157	0.99639056	2.43901563	4.13263563
393	0.049355223		0.9789441	0.99732003	2.5745165	4.3622265

5.3.5

Nonlinear equations

Energy balance conversion

$$f(X) = X - (C_p * (1 + K_A) * (T - T_c)) / (H)$$

$$X(0) = 0.80$$

Mole balance conversion

$$f(T) = X - \tau * k / (1 + \tau * k)$$

$$T(0) = 360$$

$C_p = 188.18$ # mean mixture heat capacity

$H = 66800$ # standard enthalpy of formations

$T_c = 298$ # Non-abatiatic coefficient

$\tau = 942$ # Resistance time

$E = 75730$ # activation energy

$R = 8.314$ # ideal gas constant

$k = 0.00025 * \exp((E/R) * ((1/323) - (1/T)))$ # rate constant defintion

$K_A = 3.632$ # Non-abatiatic coefficient

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